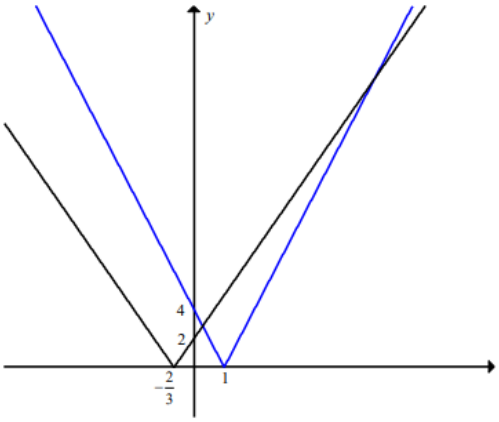
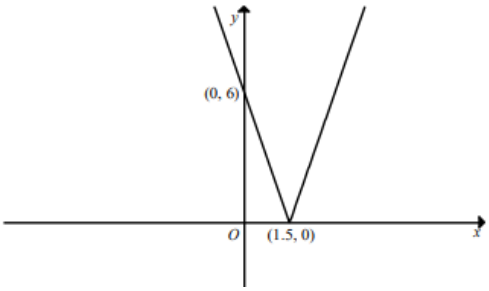
Additional Mathematics (0606)

Mark Scheme Paper 1 (Practice) — **Non-Calculator**
Chapter 1: Functions — **Sources:** CIE 0606 Past Papers
Compiler: Julian — **Quality Control**: Julian

Question	Answer	Marks	Partial Marks
1	Fully correct, ruled graphs with all intercepts indicated 	3	**M2** for a graph of correct shape with vertex on the x-axis and 1 and 4 oe appropriately indicated or a graph of correct shape with vertex on the x-axis and $-\frac{2}{3}$ and 2 oe appropriately indicated or **M1** for a graph of correct shape with vertex on the x-axis and 1 or 4 oe appropriately indicated or a graph of correct shape with vertex on the x-axis and $-\frac{2}{3}$ or 2 oe appropriately indicated
2	**(a)** $2\left(x-\frac{1}{2}\right)^2+\frac{5}{2}$ **(b)** $\frac{1}{2}$ **(c)** $f \geqslant \frac{5}{2}$	3 1 1	**B2** for $2\left(x-\frac{1}{2}\right)^2$ or **B1** for $\left(x-\frac{1}{2}\right)^2$ or $a = 2$ and $b = -0.5$ oe **B1** for $\frac{5}{2}$ or $c = 2.5$ oe **FT** *–their b* **FT** *their c*

2	**(d)** $\left[f^{-1}(x)=\right]\frac{1}{2}-\sqrt{\frac{1}{2}\left(x-\frac{5}{2}\right)}$ oe or $\left[f^{-1}(x)=\right]\frac{2-\sqrt{8x-20}}{4}$ oe	**3**	**M1 FT** for a complete method to find the inverse with a correct order of operations **FT** an expression of the form $f(x) = a(x+b)^2+c$ **A1 FT** for $\left[f^{-1}(x)=\right]\frac{1}{2}\pm\sqrt{\frac{1}{2}\left(x-\frac{5}{2}\right)}$ oe **FT** *their* $\frac{1}{2}$ and *their* $\frac{5}{2}$
3	$\frac{3\sqrt{x+2}}{\sqrt{x+2}+4}=1$ soi OR $g(x)=f^{-1}(1)$ and $f^{-1}(x)=\frac{4x}{3-x}$ oe $x=2$	**2** **2** **[T: 4]**	**M1** for $f(\sqrt{x+2})$ $[=1]$ soi or for $g(x)=f^{-1}(1)$ soi **M1 FT** for correct equation without an algebraic fraction e.g. $\sqrt{x+2}=2$ **FT** *their* $\frac{3\sqrt{x+2}}{\sqrt{x+2}+4}=1$ providing **M1** previously awarded and fg(x) is an algebraic fraction
4	Fully correct graph with intercepts marked  	**2**	**B1** for a graph of correct shape with vertex on x-axis

5	**(a)** Valid explanation: Range of g is g > 0 oe **(b)** $\dfrac{1}{\left(\dfrac{3x^2}{4x-1}\right)^2}$ $\dfrac{(4x-1)^2}{9x^4}$ or simplified equivalent, isw **(c)** $3x^2-4xy+y[=0]$ or $3y^2-4xy+x[=0]$ $[x=]\dfrac{-(-4y)\pm\sqrt{(-4y)^2-4(3)(y)}}{2(3)}$ oe or $[y=]\dfrac{-(-4x)\pm\sqrt{(-4x)^2-4(3)(x)}}{2(3)}$ oe Justifies the negative square root $f^{-1}(x)=\dfrac{2x-\sqrt{x(4x-3)}}{3}$	**1** **M1** **A1** **[T: 2]** **B1** **M1** **B1** **A1** **[T: 4]**	**FT** *their* expression providing it has at most one sign error
6	$g^2(x)=\left(\left(x^{\frac{1}{2}}-4\right)^{\frac{1}{2}}-4\right)$ $\left(\left(x^{\frac{1}{2}}-4\right)^{\frac{1}{2}}-4\right)=-2$ leading to $x^{\frac{1}{2}}=8$, $x=64$	**M1** **M1** **A1** **[T: 3]**	**Dep** on previous M mark for a correct attempt at a solution. Must deal with $x^{\frac{1}{2}}$ correctly to obtain the final solution
7	$x=\dfrac{13}{14}$ oe $7x-3=-3.5$ oe, soi or $28x-12=-14$ oe, soi $x=-\dfrac{1}{14}$ oe	**B1** **M1** **A1** **[T: 3]**	

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(a)

Expression	Function notation
0	g″
$4x$	f′
$8x^2+8x+2$	fg
$4x+3$	g^2
$\frac{x-1}{2}$	g^{-1}

(b)(i)

$a=1$

(b)(ii)

$h(x)\geqslant 3$

(b)(iii)

$x=(y-1)^2+3$

$y=1+\sqrt{x-3}$

$h^{-1}(x)=1+\sqrt{x-3}$

$x\geqslant 3$

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5

1

1

M1

A1

B1
[T: 3]

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B1 for each one correct

For a correct attempt to find the inverse, allow one sign error

Must be using correct notation

Must be using correct notation

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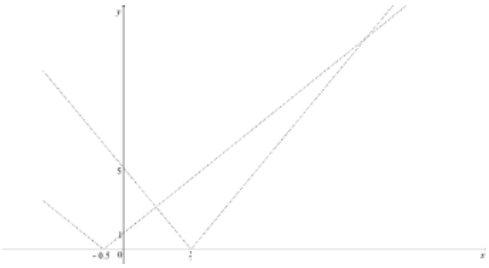

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3

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B1 for 2 V-shaped graphs with vertices in the 1st and 2nd quadrants, intersecting twice in the first quadrant.

Dep B1 for $(0,1)$ and $(0,5)$

B1 for $\left(-\frac{1}{2},0\right)$ and $\left(\frac{5}{3},0\right)$

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$f\geqslant -4$

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2

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M1for a valid method to find the least value of x^2+4x

A1 for $f\geqslant -4$, $y\geqslant -4$ or $f(x)\geqslant -4$

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(a)

Correct V shape with vertex on positive x-axis

(0, 7)

$\left(\frac{7}{5},\ 0\right)$

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B1

B1

B1
[T: 3]

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11	**(b)** $x = 2$ $5x - 7 = their(-3)$ oe, soi or $25x - 35 = their(-15)$ oe, soi $x = \frac{4}{5}$ oe **<u>Alternative Method</u>** $25x^2 - 70x + 40 = 0$ oe factorising e.g. $(5x-4)(x-2)$ $x = 2, \frac{4}{5}$	**B1** **B1** **A1** **[T: 3]** **B1** **M1** **A1** **[T: 3]**	
12	**(a)** $f > 9$ **(b)** It is a one-one function because of the restricted domain **(c)** $x = (2y+3)^2$ or equivalent $y = \frac{\sqrt{x}-3}{2}$ $f^{-1} = \frac{\sqrt{x}-3}{2}$ **(d)** $x > 9$	**1** **1** **M1** **M1** **A1** **[T: 3]** **1**	Allow y but not x For a correct attempt to find the inverse For correct rearrangement Must have correct notation **FT** on *their* (a)
13	$[\pm]\sqrt{x-1} = y - 4$ soi $g^{-1}(x) = 4 - \sqrt{x-1}$ [Range] $g^{-1} \leqslant 4$ [Domain] $x \geqslant 1$	**M1** **A1** **B1** **B1** **[T: 4]**	
14	**(a)** Many-many (mapping) oe or A vertical line meets graph more than once (or twice) oe	**1**	

14	**(b)** <table><tr><td></td><td>✓</td><td></td></tr><tr><td>✓</td><td></td><td>✓</td></tr><tr><td></td><td>✓</td><td></td></tr><tr><td>✓</td><td></td><td></td></tr></table> **(c)(i)** $\sin x \geqslant \frac{1}{2}$ $a = 150$ seen **(c)(ii)** $0 \leqslant \text{gf}(x) \leqslant \frac{1}{\sqrt{2}}$ **(c)(iii)** No. Range of g $\not\subset$ domain of g oe	**4** **M1** **A1** **[T: 2]** **2** **1**	**B1** for each correct row May be implied. Allow > **B1** for each correct value with correct sign associated with it
15	It is a one-one function because of the given restricted domain or because $x \geqslant -1$	**1**	
16	$\sqrt{2x-3}+5=7$ $x=\frac{2^2+3}{2}$ $x=\frac{7}{2}$ or 3.5	**M1** **M1** **A1** **[T: 3]**	For correct order Dep on previous M mark, for obtaining x by simplifying and solving using correct order of operations, including squaring
17	$x=3$ $2-3x=4+x$ oe $x=-0.5$ oe	**B1** **M1** **A1** **[T: 3]**	
18	**(a)(i)** Valid explanation e.g. Each x is mapped to a unique value of y [and so g is a function] but the inverse does not exist because it is many to one oe **(a)(ii)** $\left[g^2(x)=\right]\ 6(6x^4+5)^4+5$ isw for all real x **(a)(iii)** $[k=]\ 0$	**2** **2** **1**	**B1** for either each x is mapped to a unique value of y oe or for inverse does not exist because it is many to one oe **B1** for $\left[g^2(x)=\right]\ 6(6x^4+5)^4+5$ isw **B1** for correct domain

18	**(a)(iv)** $x^4 = \frac{y-5}{6}$ soi $x = \pm\sqrt[4]{\frac{y-5}{6}}$ $h^{-1}(x) = -\sqrt[4]{\frac{x-5}{6}}$	**M1** **A1** **A1** **[T: 3]**	or $y^4 = \frac{x-5}{6}$ or $y = \pm\sqrt[4]{\frac{x-5}{6}}$ If **M1 A0 A0**, allow **SC1** for an answer of $h^{-1}(x) = \sqrt[4]{\frac{x-5}{6}}$ or $y = \sqrt[4]{\frac{x-5}{6}}$
19	$g^2(x) = (x^2+4)^2 + 4$ $x^4 + 8x^2 + 16 + 4 = 40$ $(x^2+4)^2 = 36$ **or** $x^4 + 8x^2 - 20 = 0$ $(x^2+10)(x^2-2) = 0$ $x = \pm\sqrt{2}$ only	**M1** **M1** **A1** **[T: 3]**	correct use of g^2 **DepM1** for forming and solving a quadratic in x^2
20	**(a)(i)** fg **(a)(ii)** g^{-1} **(a)(iii)** f^{-1} **(a)(iv)** g^2 **(b)(i)** Undefined at $x = 0$ oe **(b)(ii)** $4 = a + b$ $h'(x) = \frac{p}{x^3}$ and attempt at $h'(1)$ $b = -8$ $a = 12$	**1** **1** **1** **1** **1** **M1** **A1** **[T: 2]**	For attempt at $h(1)$ and differentiation to obtain $h'(1)$, must have the form $h'(x) = \frac{p}{x^3}$ oe For both

21	**(i)** **(ii)** $7 = 2x - 3 \rightarrow x = 5$ Uses $7 = 3 - 2x$ oe $x = -2$	**2** **B1** **M1** **A1** **[T: 3]**	**B1** shape **B1** Correct intersection with axes.
22	**(i)** Putting $y = \mathrm{f}(x)$, changing subject to x and swopping x and y or vice versa $\mathrm{f}^{-1}(x) = \frac{1}{2}\left(\frac{1}{x} + 5\right)$ or $\frac{5x+1}{2x}$ oe isw **(ii)** $x > 0$ oe **(iii)** $\frac{1}{2\left(\frac{1}{2x-5}\right) - 5}$ $\frac{1}{\frac{2-5(2x-5)}{2x-5}}$ oe Completes to $\frac{2x-5}{-10x+27}$ oe final answer	**M1** **A1** **[T: 2]** **1** **B1** **M1** **A1** **[T: 3]**	 **FT** if expression of equivalent difficulty e.g. $\frac{1}{\left(\frac{1}{2x-5}\right) - 5}$

End of This Paper
Total Marks: 107